

DBT Lab Exam

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Q1.

Ans:

```
CREATE TABLE Category (  
    category_id INT,  
    category_name VARCHAR(50)  
);
```

```
CREATE TABLE Products (  
    product_id INT,  
    product_name VARCHAR(100),  
    price DECIMAL(10,2),  
    stock_qty INT,  
    category_id INT  
);
```

```
CREATE TABLE Sales (  
    sale_id INT,  
    product_id INT,  
    quantity INT,  
    sale_date DATE  
);
```

```
ALTER TABLE Category  
    ADD PRIMARY KEY (category_id);
```

```
ALTER TABLE Products  
    ADD PRIMARY KEY (product_id);
```

```
ALTER TABLE Sales  
    ADD PRIMARY KEY (sale_id);
```

```
ALTER TABLE Products  
    ADD CONSTRAINT fk_products_category  
    FOREIGN KEY (category_id) REFERENCES Category(category_id);
```

```
ALTER TABLE Sales  
    ADD CONSTRAINT fk_sales_product  
    FOREIGN KEY (product_id) REFERENCES Products(product_id);
```

```
INSERT INTO Category (category_id, category_name) VALUES  
(1, 'Grocery'),  
(2, 'Dairy'),  
(3, 'Bakery');
```

```
INSERT INTO Products (product_id, product_name, price, stock_qty, category_id) VALUES  
(101, 'Basmati Rice 5kg', 450.00, 50, 1),
```

```
(102, 'Amul Milk 1L', 60.00, 100, 2),  
(103, 'Brown Bread', 45.00, 30, 3);
```

```
INSERT INTO Sales (sale_id, product_id, quantity, sale_date) VALUES  
(1, 101, 2, '2026-07-01'),  
(2, 102, 5, '2026-07-02'),  
(3, 103, 1, '2026-07-03');
```

```
SELECT * FROM Products  
ORDER BY product_name ASC;
```

```
SELECT * FROM Products  
WHERE category_id = 1 OR category_id = 2;
```

```
SELECT * FROM Products  
LIMIT 5;
```

```
SELECT * FROM Products  
WHERE price BETWEEN 40 AND 500;
```

```
SELECT category_id, COUNT(*) AS total_products, SUM(stock_qty) AS total_stock  
FROM Products  
GROUP BY category_id;
```

```
SELECT Products.product_name, Sales.quantity, Sales.sale_date  
FROM Products  
LEFT JOIN Sales ON Products.product_id = Sales.product_id;
```

```
CREATE TEMPORARY TABLE Temp_Products AS  
SELECT * FROM Products WHERE stock_qty > 20;
```

```
SELECT * FROM Temp_Products;
```

```
DROP TEMPORARY TABLE Temp_Products;
```

```
DELIMITER //
```

```
CREATE PROCEDURE GetStockByProduct(  
    IN p_product_id INT,  
    OUT p_stock INT  
)  
BEGIN  
    SELECT stock_qty INTO p_stock  
    FROM Products  
    WHERE product_id = p_product_id;  
END //
```

```
DELIMITER ;
```

```
-- To call the procedure:  
CALL GetStockByProduct(101, @stock);
```

```
SELECT @stock;
```

```
DELIMITER //
```

```
CREATE TRIGGER before_product_update
```

```
BEFORE UPDATE ON Products
```

```
FOR EACH ROW
```

```
BEGIN
```

```
    IF NEW.stock_qty < 0 THEN
```

```
        SET NEW.stock_qty = 0;
```

```
    END IF;
```

```
END //
```

```
DELIMITER ;
```

product_id	product_name	price	stock_qty	category_id
102	Amul Milk 1L	60.00	100	2
101	Basmati Rice 5kg	450.00	50	1
103	Brown Bread	45.00	30	3
NULL	NULL	NULL	NULL	NULL

product_id	product_name	price	stock_qty	category_id
101	Basmati Rice 5kg	450.00	50	1
102	Amul Milk 1L	60.00	100	2
NULL	NULL	NULL	NULL	NULL

product_id	product_name	price	stock_qty	category_id
101	Basmati Rice 5kg	450.00	50	1
102	Amul Milk 1L	60.00	100	2
103	Brown Bread	45.00	30	3
NULL	NULL	NULL	45.00	NULL

product_id	product_name	price	stock_qty	category_id
101	Basmati Rice 5kg	450.00	50	1
102	Amul Milk 1L	60.00	100	2
103	Brown Bread	45.00	30	3
NULL	NULL	NULL	NULL	NULL

category_id	total_products	total_stock
1	1	50
2	1	100
3	1	30

Result Grid			Filter Rows:	Export: 	Wrap Cell Content: 
	product_name	quantity	sale_date		
▶	Basmati Rice 5kg	2	2026-07-01		
	Amul Milk 1L	5	2026-07-02		
	Brown Bread	1	2026-07-03		

Result Grid			Filter Rows:	Export: 	Wrap Cell Content: 
	product_id	product_name	price	stock_qty	category_id
▶	101	Basmati Rice 5kg	450.00	50	1
	102	Amul Milk 1L	60.00	100	2
	103	Brown Bread	45.00	30	3

Result Grid			Filter Rows:	Export: 	Wrap Cell Content: 
	@stock				
▶	50				

Q2.

Ans:

use supermarket

```
db.createCollection("products")
```

```
db.products.insertMany([
  { product_id: 101, product_name: "Basmati Rice 5kg", price: 450, category: "Grocery", stock_qty: 50 },
  { product_id: 102, product_name: "Amul Milk 1L", price: 60, category: "Dairy", stock_qty: 100 },
  { product_id: 103, product_name: "Brown Bread", price: 45, category: "Bakery", stock_qty: 30 }
])
```

```
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a70ef0f6139de6de3747b00'),
    '1': ObjectId('6a70ef0f6139de6de3747b01'),
    '2': ObjectId('6a70ef0f6139de6de3747b02')
  }
}
```

```
db.products.find()
```

```
> db.products.find()
< {
  _id: ObjectId('6a70ef0f6139de6de3747b00'),
  product_id: 101,
  product_name: 'Basmati Rice 5kg',
  price: 450,
  category: 'Grocery',
  stock_qty: 50
}
{
  _id: ObjectId('6a70ef0f6139de6de3747b01'),
  product_id: 102,
  product_name: 'Amul Milk 1L',
  price: 60,
  category: 'Dairy',
  stock_qty: 100
}
{
  _id: ObjectId('6a70ef0f6139de6de3747b02'),
  product_id: 103,
  product_name: 'Brown Bread',
  price: 45,
  category: 'Bakery',
  stock_qty: 30
}
```

```
db.products.find({
  $and: [
    { price: { $gt: 100 } },
    { category: "Grocery" }
  ]
})
```

```
< {
  _id: ObjectId('6a70ef0f6139de6de3747b00'),
  product_id: 101,
  product_name: 'Basmati Rice 5kg',
  price: 450,
  category: 'Grocery',
  stock_qty: 50
}
```

```
db.products.find().sort({ price: -1 })
```

```
< {
  _id: ObjectId('6a70ef0f6139de6de3747b00'),
  product_id: 101,
  product_name: 'Basmati Rice 5kg',
  price: 450,
  category: 'Grocery',
  stock_qty: 50
}
{
  _id: ObjectId('6a70ef0f6139de6de3747b01'),
  product_id: 102,
  product_name: 'Amul Milk 1L',
  price: 60,
  category: 'Dairy',
  stock_qty: 100
}
{
  _id: ObjectId('6a70ef0f6139de6de3747b02'),
  product_id: 103,
  product_name: 'Brown Bread',
  price: 45,
  category: 'Bakery',
  stock_qty: 30
}
```

```
db.products.createIndex({ product_name: 1 })
```